

JEE Mains 2019 Chapter wise Question Bank

Nuclear Physics - Questions

Q1

A sample of radioactive material A, that has an activity of 10 mCi ($1 \text{ Ci} = 3.7 \times 10^{10} \text{ decays/s}$), has twice the number of nuclei as another sample of a different radioactive material B which has an activity of 20 mCi. The correct choices for half-lives of A and B would then be respectively:

- (1) 5 days and 10 days (2) 10 days and 40 days
(3) 20 days and 5 days (4) 20 days and 10 days

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Q2

At a given instant, say $t = 0$, two radioactive substances

A and B have equal activities. The ratio $\frac{R_B}{R_A}$ of their activities after time t itself decays with time t as e^{-3t} . If the half-life of A is $\ln 2$, the half-life of B is:

- (1) $4\ln 2$ (2) $\frac{\ln 2}{2}$
(3) $\frac{\ln 2}{4}$ (4) $2\ln 2$

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Q3

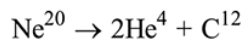
Using a nuclear counter the count rate of emitted particles from a radioactive source is measured. At $t = 0$ it was 1600 counts per second and $t = 8$ seconds it was 100 counts per second. The count rate observed, as counts per second, at $t = 6$ seconds is close to:

- (1) 200 (2) 150 (3) 400 (4) 360

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Q4

Consider the nuclear fission



Given that the binding energy/nucleon of Ne^{20} , He^4 and C^{12} are, respectively, 8.03 MeV, 7.07 MeV and 7.86 MeV, identify the correct statement:

- (1) energy of 12.4 MeV will be supplied
(2) 8.3 MeV energy will be released
(3) energy of 3.6 MeV will be released
(4) energy of 11.9 MeV has to be supplied

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Q5

In a radioactive decay chain, the initial nucleus is ${}_{90}^{232}\text{Th}$.

At the end there are 6 α -particles and 4 β -particles which are emitted. If the end nucleus is ${}_Z^AX$, A and Z are given by:

- (1) $A = 208$; $Z = 80$ (2) $A = 202$; $Z = 80$
(3) $A = 208$; $Z = 82$ (4) $A = 200$; $Z = 81$

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Q6

The ratio of the mass densities of nuclei of ${}^{40}\text{Ca}$ and ${}^{16}\text{O}$ is close to:

- (1) 1 (2) 0.1 (3) 5 (4) 2

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Q7

A nucleus A, with a finite de-broglie wavelength λ_A , undergoes spontaneous fission into two nuclei B and C of equal mass. B flies in the same direction as that of A, while C flies in the opposite direction with a velocity equal to half of that of B. the de-Broglie wavelengths λ_B and λ_C of B and C are respectively:

- (1) $\lambda_A, 2\lambda_A$ (2) $2\lambda_A, \lambda_A$ (3) $\lambda_A, \frac{\lambda_A}{2}$ (4) $\frac{\lambda_A}{2}, \lambda_A$

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Q8

Two radioactive materials A and B have decay constants 10λ and λ , respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of A to that of B will be $1/e$ after a time :

- (1) $\frac{1}{9\lambda}$ (2) $\frac{1}{11\lambda}$ (3) $\frac{11}{10\lambda}$ (4) $\frac{1}{10\lambda}$

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Q9

Two radioactive substances A and B have decay constants 5λ and λ respectively. At $t = 0$, a sample has the same number of the two nuclei. The time taken for the ratio of the

number of nuclei to become $\left(\frac{1}{e}\right)^2$ will be :

- (1) $1/2\lambda$ (2) $1/4\lambda$ (3) $1/\lambda$ (4) $2/\lambda$

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Q10

Half lives of two radioactive nuclei A and B are 10 minutes and 20 minutes, respectively. If, initially a sample has equal number of nuclei, then after 60 minutes, the ratio of decayed numbers of nuclei A and B will be :

- (1) 3 : 8 (2) 1 : 8 (3) 8 : 1 (4) 9 : 8

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Nuclear Physics - Answers

Q1

$$\begin{aligned} \text{For material, A} \quad 10 &= (2 N_0) \lambda_A \\ \text{For material, B} \quad 20 &= N_0 \lambda_B \end{aligned}$$

$$\Rightarrow \lambda_B = 4\lambda_A \therefore T_{1/2A} = 4T_{1/2B} \left[\because T_{1/2} = \frac{0.693}{\lambda} \right]$$

i.e. 20 days half-lives for A and 5 days $(T_{1/2})_B$ For material B.

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Q2

(3) Half life of A = $\ell \ln 2$

$$(t_{1/2})_A = \frac{\ell \ln 2}{\lambda}$$

$$\therefore \lambda_A = 1$$

$$\text{at } t = 0 \quad R_A = R_B$$

$$N_A e^{-\lambda_A t} = N_B e^{-\lambda_B t}$$

$$N_A = N_B \text{ at } t = 0$$

$$\text{At } t = t \quad \frac{R_B}{R_A} = \frac{N_0 e^{-\lambda_B t}}{N_0 e^{-\lambda_A t}}$$

$$e^{-(\lambda_B - \lambda_A)t} = e^{-3t}$$

$$\Rightarrow \lambda_B - \lambda_A = 3$$

$$\lambda_B = 3 + \lambda_A = 4$$

$$(t_{1/2})_B = \frac{\ell \ln 2}{\lambda_B} = \frac{\ell \ln 2}{4}$$

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Q3

(1) According to question, at $t = 0$, $A_0 = \frac{dN}{dt} = 1600 \text{ C/s}$
and at $t = 8\text{s}$, $A = 100 \text{ C/s}$

$$\therefore \frac{A}{A_0} = \frac{1}{16} \text{ in } 8\text{s}$$

Therefore half life period, $t_{1/2} = 2\text{s}$

$$\therefore \text{Activity at } t = 6\text{s} = 1600 \left(\frac{1}{2}\right)^3 = 200 \text{ C/s}$$

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Q4



$$8.03 \times 20 \quad 2 \times 7.07 \times 4 + 7.86 \times 12$$

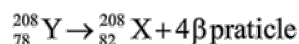
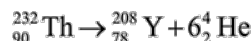
$$\therefore E_B = (BE)_{\text{react}} - (BE)_{\text{product}} = 9.72 \text{ MeV}$$

therefore, energy has to be supplied

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Q5

(3) When one α -particle emitted then daughter nuclei has 4 unit less mass number (A) and 2 unit less atomic (z) number (z).



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Q6

(1) Nuclear density is independent of atomic number.

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Q7

Nuclear Physics

$$\begin{aligned}
 (4) \quad \vec{P} &= \vec{P} + \vec{P}_2 \\
 \text{or } \vec{P} &= \vec{P}' - \frac{\vec{P}'}{2} \\
 &= \frac{\vec{P}'}{2} \\
 \text{As } \lambda &= \frac{h}{P} \\
 \Rightarrow \lambda_B &= \frac{\lambda_A}{2} \text{ and } \lambda_C = \lambda_A
 \end{aligned}$$

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Q8

$$\begin{aligned}
 (1) \text{ As, } N &= N_0 e^{-\lambda t} \\
 \text{so, } \frac{N_A}{N_B} &= e^{(\lambda_B - \lambda_A)t} = \frac{1}{e} \Rightarrow (\lambda_B - \lambda_A)t = -1 \\
 \Rightarrow (\lambda_A - \lambda_B) \cdot t &= 1 \\
 \Rightarrow t &= \frac{1}{(\lambda_B - \lambda_A)} \quad t = \frac{1}{10\lambda - \lambda} = \frac{1}{9\lambda}
 \end{aligned}$$

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Q9

(1) Let N_1 and N_2 be the number of radioactive nuclei of substance at anytime t .

$$N_1(\text{at } t) = N_0 e^{-5\lambda t} \quad (i)$$

$$N_2(\text{at } t) = N_0 e^{-\lambda t} \quad (ii)$$

Dividing equation (i) by (ii), we get

$$\frac{N_1}{N_2} = \frac{1}{e^2} = e^{-4\lambda t}$$

$$\Rightarrow 4\lambda t = 2$$

$$\Rightarrow t = \frac{2}{4\lambda} = \left(\frac{1}{2\lambda} \right)$$

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Q10

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$$\begin{aligned}
 (4) \quad \frac{N_1}{N_2} &= \frac{N_0(1 - e^{-\lambda_1 t})}{N_0(1 - e^{-\lambda_2 t})} \\
 \text{or } \frac{N_1}{N_2} &= \frac{\left(1 - e^{-\frac{0.693}{10} \times 60} \right)}{\left(1 - e^{-\frac{0.693}{20} \times 60} \right)} = \frac{\left(1 - e^{-4.2} \right)}{\left(1 - e^{-2.1} \right)} = \frac{9}{8}
 \end{aligned}$$

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